

SENIOR PROJECT CN1-2025

Gait Recognition

Project Concept

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School of Information, Computer and Communication Technology
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By

Satayu Satiraungkul 6522770674
Ravipas Panutatpinyo 6522771003
Nattapat Punmanee 6522771144

Advisor : Dr. Cholwich Nattee

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Project Overview

Abstract

Gait recognition is a biometric method that identifies individuals based on their walking patterns, offering a non-intrusive alternative to traditional identification systems. This study presents the development of a gait recognition system that progresses from a dataset-based model to a real-time application.

An initial approach utilized a large-scale dataset containing 3D skeleton data and a Spatio-Temporal Graph Convolutional Network (ST-GCN), achieving strong performance under controlled conditions. However, a mismatch between 3D training data and 2D keypoints from real-time pose estimation prevented effective real-world use .

To address this issue, a new dataset was collected using video recordings, and skeleton keypoints were extracted using YOLO-Pose. The system was redesigned to ensure consistency between training and inference data. A complete pipeline was implemented for real-time recognition, including preprocessing, model training, and live inference with a confidence-based unknown detection mechanism.

The final system achieved an accuracy of 87.5% on the self-collected dataset and demonstrated stable real-time performance. The results highlight the importance of data consistency and show that gait recognition can be effectively applied in practical environments.

1. Introduction

Biometric identification systems are widely used in security, authentication, and surveillance systems. Common modalities include facial recognition, fingerprint scanning, iris recognition, and voice recognition. However, these methods require close-range capture or active user participation .

Gait recognition differs because it analyzes walking motion, which is a natural and unconscious behavior. It allows identification at a distance using standard cameras, making it a non-intrusive method.

Human gait is influenced by:

- Bone structure
- Muscle dynamics
- Body proportions
- Neuromuscular coordination

These factors create a unique motion pattern for each individual.

Earlier gait recognition systems relied on silhouette-based or 3D mesh-based approaches. While accurate, these methods are either sensitive to environmental conditions or computationally expensive. With recent advancements in pose estimation and graph neural networks, it is now possible to develop efficient and real-time gait recognition systems.

This study aims to design and implement a modern gait recognition system that transitions from a dataset-based model to a practical real-time application.

1.1 Contribution of This Work

This work contributes to the development of a practical gait recognition system that operates in real-world conditions. Unlike traditional approaches that rely on pre-collected datasets, this system ensures consistency between training and real-time input by using self-collected data and pose estimation. The proposed system integrates YOLO-Pose and ST-GCN to achieve real-time performance and includes an unknown detection mechanism for improved reliability.

2. Literature Review

2.1 Silhouette-Based Methods

Early methods use Gait Energy Image (GEI), which averages silhouette images over time. These methods are simple but sensitive to noise, clothing, and lighting .

2.2 Mesh-Based Methods

The OUMVLP-Mesh dataset provides detailed 3D human body representations. While accurate, these methods require high computational resources and are difficult to deploy in real-time systems .

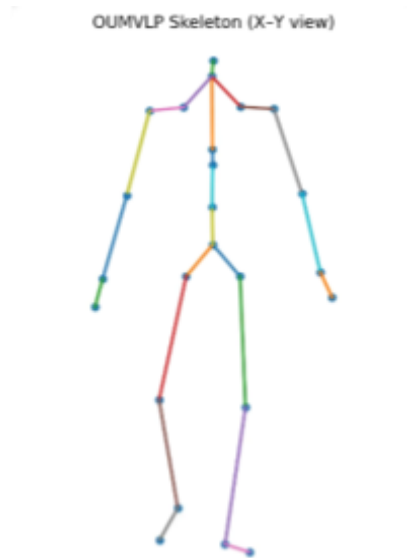


Figure 1: Example of 3D skeletal joint representation from the OUMVLP-Mesh dataset (XY view).

2.3 Skeleton-Based Deep Learning

Skeleton-based methods represent human motion as joint coordinates. This approach:

- Reduces computational complexity
- Focuses on motion patterns
- Is robust to appearance variation

ST-GCN models the skeleton as a graph:

- Nodes = joints
- Edges = bone connections
- Time dimension = motion

The spatial graph convolution is defined as:

$$\mathbf{A} \cdot \mathbf{X} \cdot \mathbf{W}$$

Where:

- A = adjacency matrix
- X = input joint features
- W = learnable weights

This architecture is effective for gait recognition.

3. Problem Statement

Existing gait recognition systems face several limitations:

- Silhouette-based methods lack robustness
- Mesh-based methods are computationally expensive
- Many systems are not suitable for real-time deployment

Additionally, a critical issue arises when applying dataset-trained models to real-world scenarios. Training data often uses structured formats such as 3D skeletons, while real-time systems generate 2D pose keypoints.

This mismatch prevents the model from correctly interpreting real-world input, leading to unreliable predictions .

Therefore, a system is required that:

- Supports real-time inference
- Uses lightweight computation
- Maintains consistent data representation
- Accurately identifies known users and rejects unknown users

4. System Overview

The proposed system follows an end-to-end pipeline:

1. Video collection
2. Pose extraction using YOLO-Pose
3. Skeleton format conversion
4. Data normalization
5. Dataset construction
6. Model training using ST-GCN
7. Real-time inference

This design ensures consistency between training data and real-time input.

5. Dataset and Data Preparation

To resolve the incompatibility issue, a new dataset was constructed using self-collected video recordings. The dataset consists of walking sequences recorded from multiple subjects, with each video lasting approximately 3 to 9 seconds at 60 frames per second.

The dataset is organized as follows:

- Subject A(It): 22 walking clips (18 training, 4 testing)
- Subject B(Ment): 19 walking clips (15 training, 4 testing)
- Total: 41 clips (33 training, 8 testing)

Subject	Walking clips	Train clips	Test clips
It	22	18	4
Ment	19	15	4
Total	41	33	8

All videos are stored in a structured directory format and processed through a multi-stage pipeline.

5.1 Pose Extraction

Each video is processed using YOLO-Pose to extract 2D keypoints for every frame. The output consists of 17 keypoints with associated confidence values.

These keypoints are then converted into a 24-joint skeleton format to match the model structure. Additional joints such as pelvis, neck, and spine are computed using midpoint calculations between existing keypoints.

5.2 Data Normalization

A shared preprocessing module is used to ensure consistency across training and inference. This module performs:

- Centering of skeleton based on the pelvis joint
- Scaling using average spine length
- Handling missing joints by zeroing low-confidence keypoints

This ensures that the model learns motion patterns rather than absolute positions or scale.

5.3 Format Consistency

A critical observation was made during data collection. When mixing

horizontal and vertical video formats, model performance degraded significantly, with accuracy dropping from 87.5% to 55.6%.

Further analysis revealed that differences in aspect ratio caused large variations in keypoint distribution, making samples of the same subject appear more different than samples of different subjects.

After standardizing all recordings to a consistent horizontal format, model performance recovered. This demonstrates that format consistency is essential for reliable model performance.

6. System Architecture

The final system consists of an end-to-end pipeline with the following stages:

1. Video collection
2. Pose extraction using YOLO-Pose
3. Conversion to standardized skeleton format
4. Data normalization
5. Dataset construction
6. Model training using ST-GCN
7. Real-time inference

6.1 Preprocessing Module

A shared preprocessing module is implemented to ensure identical transformations are applied during both training and inference. This eliminates distribution mismatch between training data and real-time input.

6.2 ST-GCN Model

The ST-GCN model is implemented with a proper graph convolution

structure:

- Adjacency matrix representing skeletal connections
- Graph convolution layers for spatial relationships
- Temporal convolution layers for motion patterns
- Batch normalization and ReLU activation
- Residual connections for stable training

The model consists of three main blocks with increasing channel dimensions ($64 \rightarrow 128 \rightarrow 256$), followed by global pooling and a fully connected classification layer.

7. Training Methodology

In this redesigned system, a new ST-GCN model was trained using the self-collected dataset to ensure compatibility with real-time input.

Unlike the initial approach, which relied on 3D skeleton data, the new model is trained entirely on 2D keypoints extracted using YOLO-Pose.

Several improvements were introduced to ensure stable training on a small dataset:

- Stratified train/test split to maintain class balance
- Class weighting to address data imbalance
- Data augmentation including:
 - horizontal flipping
 - temporal cropping and resizing
 - coordinate jitter
- Cosine learning rate scheduling for gradual convergence
- Early stopping to prevent overfitting

These techniques improve generalization despite limited training data.

8. Implementation

8.1 Structure

The system is implemented as a modular architecture consisting of separate components for data processing, training, and real-time inference.

During real-time operation:

- Frames are captured from a webcam
- YOLO-Pose extracts keypoints per frame
- A sliding buffer of 75 frames is maintained
- Predictions are generated every 5 frames

A confidence-based classification system is used:

- Confidence $\geq 60\%$ $\rightarrow$ display predicted identity
- Confidence $< 60\%$ $\rightarrow$ display “Unknown”

A visual overlay provides real-time feedback, including prediction results and buffer status.

8.1 Updated Configuration

Parameter	Old	New	Reason
Batch size	16	4	More gradient updates
Epochs	30	80	Slower learning at low learning rates
Learning rate	5e-4	3e-4	More stable on tiny data
Optimizer	Adam	Adam + weight decay 1e-4	Mild regularisation
LR schedule	constant	Cosine annealing	Stabilises later epochs
Sequence length	75 frames	75 frames	Unchanged

9. Results

9.1 Experimental Setup

- **Hardware:** Standard laptop, Intel CPU only (no GPU used for inference).
- **Camera:** Built-in webcam, horizontal orientation, 720p.
- **Dataset:** 41 self-collected clips (22 It, 19 Ment) at 60 fps.
- **Train/test split:** 80/20 stratified, deterministic with seed 42.

9.2 Quantitative Results

The model achieved the following performance:

- Subject A: 100% accuracy (4/4 correct)
- Subject B: 75% accuracy (3/4 correct)
- Overall accuracy: 87.5% (7/8 correct)

Class	Test samples	Correct	Accuracy
[0] It	4	4	100.0%
[1] Ment	4	3	75.0%
Overall	8	7	87.5%

A confusion matrix analysis shows minor misclassification between subjects.

	Pred It	Pred Ment
True It	4	0
True Ment	1	3

9.3 Real-Time Performance

In live testing:

- **It**: recognised correctly with confidence in the 60-90% range.
- **Ment**: recognised correctly with confidence in the 70-90% range.
- **Strangers (not enrolled)**: predictions oscillate between It and Ment with confidence usually 50-70%, frequently triggering the "Unknown" threshold.

The full 75-frame buffer fills in approximately 1-2 seconds of walking. After that, predictions update at roughly 6 Hz.

9.4 Comparison

Aspect	Planned	Actual
Training data	OUMVLP-Mesh (3D, 11 subjects)	Self-collected (2D, 2 subjects)
Test accuracy	93.14% (on OUMVLP)	87.5% (on self-collected)
Real-time demo	Mock-up only	Working webcam application
Train/inference consistency	Inconsistent (3D train, 2D infer)	Consistent (2D throughout)
Unknown detection	Specified ("Not match")	Implemented (60% threshold, red overlay)

“The decrease in accuracy from 93.14% to 87.5% reflects the increased difficulty of real-world data compared to controlled dataset conditions.”

10. Discussion

The results emphasize the importance of matching training and inference data distributions. The initial dataset-based approach achieved high accuracy but failed in real-world scenarios due to format mismatch.

The redesigned system demonstrates that smaller datasets can produce

reliable performance when data representation is consistent.

Additionally, the study highlights the impact of recording conditions, preprocessing methods, and data consistency on model performance.

11. Limitations

The system currently has several limitations:

- Limited number of subjects
- Single recording environment
- Dependence on 2D pose estimation
- Heuristic-based unknown detection

12. Future Work

Future improvements include:

- Expanding dataset size and diversity
- Testing across multiple environments
- Using embedding-based recognition methods
- Supporting multi-person tracking
- Improving prediction stability

13. Conclusion

This work presents a complete gait recognition system that transitions from a dataset-based approach to a real-time application. By addressing data format mismatch and ensuring consistency between training and inference, the system achieves both functional accuracy and practical usability.

The results demonstrate that skeleton-based gait recognition can be effectively applied in real-world scenarios using standard camera input.

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